Chapter 9, Problem 25P

Problem

Using phasors, determine $i(t)$ in the following equations:

(a) $2\frac{di}{dt} + 3i(t) = 4\cos(2t - 45°)$

(b) $10\int i\,dt + \frac{di}{dt} + 6i(t) = 5\cos(5t + 22°)$ A

Step-by-step solution

Step 1 of 4

(a)

The differential equation is,

$$2\frac{di}{dt} + 3i(t) = 4\cos(2t - 45°)$$

The value of angular frequency is, $\omega = 2$ rad/s

Convert the given time domain expression into phasor domain.

$$2j\omega\mathbf{I} + 3\mathbf{I} = 4\angle -45°$$

Step 2 of 4

Substitute 2 rad/s for ω.

$$2j(2)\mathbf{I} + 3\mathbf{I} = 4\angle -45°$$
$$\mathbf{I}(3 + j4) = 4\angle -45°$$
$$\mathbf{I} = \frac{4\angle -45°}{5\angle 53.13°} = 0.8\angle -98.13°$$

Convert the value of current, $\mathbf{I}$ from phasor domain to time domain.

$i(t) = 0.8\cos(2t - 98.13°)$ A Therefore, the value of $i(t)$ is $\boxed{0.8\cos(2t - 98.13°)\ \text{A}}$.

Step 3 of 4

(b)

The integrodifferential equation is,

$$10\int i\,dt + \frac{di}{dt} + 6i(t) = 5\cos(5t + 22°)$$

The value of angular frequency is, $\omega = 5$ rad/s

Convert the given time domain expression into phasor domain.

$$\frac{10\mathbf{I}}{j\omega} + j\omega\mathbf{I} + 6\mathbf{I} = 5\angle 22°$$

Step 4 of 4

Substitute 5 rad/s for ω.

$$\frac{10\mathbf{I}}{j5}+j5\mathbf{I}+6\mathbf{I}=5\angle 22^\circ$$
$$-j2\mathbf{I}+j5\mathbf{I}+6\mathbf{I}=5\angle 22^\circ$$
$$(6+j3)\mathbf{I}=5\angle 22^\circ$$
$$\begin{aligned}\mathbf{I}&=\frac{5\angle 22^\circ}{6.708\angle 26.565^\circ}\\&=0.745\angle -4.56^\circ\end{aligned}$$

Convert the value of current, $\mathbf{I}$ from phasor domain to time domain.

$i(t)=0.745\cos(5t-4.56^\circ)\text{ A}$ Therefore, the value of $i(t)$ is $\boxed{0.745\cos(5t-4.56^\circ)\text{ A}}$.